Chapter 7. Finite Fourier Analysis

Fourier analysis on Z(N) and finite abelian groups

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7.1 Fourier analysis on $\mathbb{Z}(N)$

Motivation. On the N-th roots of unity Fourier series are finite sums with an orthogonal basis of exponentials, and even/odd splitting computes them in $O(N \log N)$ (FFT).

7.1.1 The group $\mathbb{Z}(N)$

For $N \geq 1$ let $\zeta = e^{2\pi i/N}$. The roots $\{1, \zeta, \dots, \zeta^{N-1}\}$ form a multiplicative group, identified via $\zeta^k \mapsto k$ with $\mathbb{Z}(N) = \mathbb{Z}/N\mathbb{Z}$; functions on $\mathbb{Z}(N)$ are N-periodic functions on $\mathbb{Z}$, or functions on N equally spaced points of the circle (*discretization*).

7.1.2 Fourier inversion theorem and Plancherel identity on $\mathbb{Z}(N)$

The functions on $\mathbb{Z}(N)$ form an N-dimensional space V with $(F, G) = \sum_{k=0}^{N-1} F(k)\overline{G(k)}$, $\|F\|^2 = \sum |F(k)|^2$. Let $e_\ell(k) = \zeta^{\ell k}$, $\ell = 0, \dots, N-1$.

Lemma 1.1

The family $\{e_0, \dots, e_{N-1}\}$ is orthogonal: $(e_m, e_\ell) = N$ if $m = \ell$ and 0 if $m \neq \ell$. Hence $\{e_\ell/\sqrt{N}\}$ is an orthonormal basis of V.

Proof. $(e_m, e_\ell) = \sum_{k=0}^{N-1} \zeta^{(m-\ell)k}$, which is N if $m = \ell$; otherwise $q = \zeta^{m-\ell} \neq 1$, $q^N = 1$ and the sum is $\frac{1-q^N}{1-q} = 0$. N orthogonal nonzero vectors form a basis. ■

The n-th **Fourier coefficient** of F is $a_n = \frac{1}{N}(F, e_n) = \frac{1}{N}\sum_{k=0}^{N-1} F(k)\zeta^{-kn}$.

Theorem 1.2

If F is a function on $\mathbb{Z}(N)$, then $F(k) = \sum_{n=0}^{N-1} a_n \zeta^{nk}$ (inversion) and $\sum_{n=0}^{N-1} |a_n|^2 = \frac{1}{N}\sum_{k=0}^{N-1} |F(k)|^2$ (Parseval).

Proof. In the orthonormal basis $e_n^* = e_n/\sqrt{N}$: $F = \sum (F, e_n^*)e_n^* = \sum a_n e_n$, and Pythagoras gives $\|F\|^2 = \sum N|a_n|^2$. ■

(If $F(k) = f(k/N)$ for 1-periodic f, a_n is a Riemann sum for $\hat{f}(n)$.)

7.1.3 The fast Fourier transform

Naively all a_n cost about N^2 operations. Let $\#(M)$ be the least number of operations needed for all Fourier coefficients on $\mathbb{Z}(M)$, with $\omega_M = e^{-2\pi i/M}$ and $a_k^M(F) = \frac{1}{M}\sum_r F(r)\omega_M^{kr}$.

Theorem 1.3

For $N = 2^n$, the Fourier coefficients of a function on $\mathbb{Z}(N)$ can be computed with at most $4 \cdot 2^n n = 4N \log_2 N = O(N \log N)$ operations, given ω_N.

Lemma 1.4

Given ω_{2M}, $\#(2M) \leq 2\#(M) + 8M$.

Proof. $\omega_{2M},\dots,\omega_{2M}^{2M}$ cost $\le 2M$ operations and give $\omega_M=\omega_{2M}^2$. With $F_0(r)=F(2r)$, $F_1(r)=F(2r+1)$ on $\mathbb{Z}(M)$, splitting into even and odd indices,

$$\begin{aligned} a_k^{2M}(F) &= \frac{1}{2M}\sum_{r=0}^{2M-1} F(r)\omega_{2M}^{kr} = \frac{1}{2}\Big(\frac{1}{M}\sum_{\ell} F_0(\ell)\omega_M^{k\ell} + \frac{1}{M}\sum_{m} F_1(m)\omega_M^{km}\omega_{2M}^{k}\Big) \\ &= \frac{1}{2}\big(a_k^M(F_0) + a_k^M(F_1)\omega_{2M}^k\big) \end{aligned}$$

(a_k^M M-periodic in k). $a^M(F_0)$, $a^M(F_1)$ cost $2\#(M)$, then 3 operations per coefficient: $\#(2M)\le 2M+2\#(M)+6M$. ∎

Proof of Theorem 1.3. Induction on n. $N=2$: $a_0=\frac{1}{2}(F(0)+F(1))$, $a_1=\frac{1}{2}(F(0)-F(1))$, ≤ 8 operations. If $\#(2^{n-1})\le 4\cdot 2^{n-1}(n-1)$, Lemma 1.4 gives $\#(2^n)\le 2\cdot 4\cdot 2^{n-1}(n-1)+8\cdot 2^{n-1}=4\cdot 2^n n$. ∎

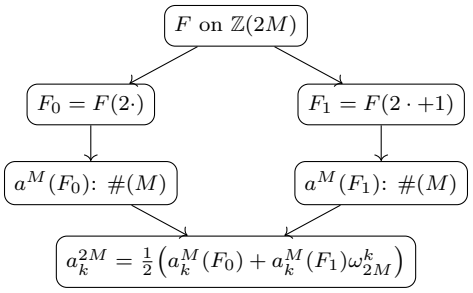


Figure. *One step of the fast Fourier transform (Lemma 1.4).*

7.2 Fourier analysis on finite abelian groups

Motivation. On a finite abelian group, characters replace exponentials; they are orthonormal and, by simultaneous diagonalization of translations, a basis.

7.2.1 Abelian groups

An **abelian group** $(G,\cdot)$ is associative and commutative with identity 1 and inverses: e.g. $\mathbb{Z}(N)$, $\mathbb{R}/\mathbb{Z}$, $\mathbb{R}$, $\mathbb{R}^+=(0,\infty)$ under multiplication, and $\mathbb{Z}^*(q)$, the residues coprime to q under multiplication. A **homomorphism** satisfies $f(ab)=f(a)f(b)$; an **isomorphism** is a bijective one. $G_1\times G_2$ multiplies componentwise ($\mathbb{Z}^*(8)\cong\mathbb{Z}(2)\times\mathbb{Z}(2)$, $\mathbb{Z}^*(5)\cong\mathbb{Z}(4)$). Every finite abelian group is a product of cyclic groups; we will not need this.

7.2.2 Characters

A **character** of G is a homomorphism $e: G\to S^1=\{|z|=1\}$; the **trivial character** is $e\equiv 1$, and $\hat{G}$ is the set of characters. On $\mathbb{Z}(N)$ these are exactly $e_\ell(k)=\zeta^{\ell k}$ ($e(1)$ is an N-th root of unity).

Lemma 2.1

$\hat{G}$ is an abelian group under $(e_1e_2)(a)=e_1(a)e_2(a)$.

Proof. Products of characters are characters, the trivial one is the identity, $e^{-1}=\bar{e}$ ($|e|=1$), and the laws hold pointwise. ∎

Lemma 2.2

If G is finite and $e : G \to \mathbb{C} - \{0\}$ satisfies $e(ab) = e(a)e(b)$, then e is a character.

Proof. If $|e(a)| > 1$ then $|e(a^n)| = |e(a)|^n \to \infty$, and if $|e(a)| < 1$ then $|e(a^n)| \to 0$; both impossible as G is finite. So $|e(a)| = 1$. ■

7.2.3 The orthogonality relations

On functions on finite G use $(f, g) = \frac{1}{|G|} \sum_{a \in G} f(a)\overline{g(a)}$.

Theorem 2.3

The characters of G form an orthonormal family in V.

Lemma 2.4

If e is a nontrivial character, then $\sum_{a \in G} e(a) = 0$.

Proof. Take $e(b) \neq 1$. As $a \mapsto ba$ permutes G, $e(b) \sum_a e(a) = \sum_a e(ba) = \sum_a e(a)$, so $\sum_a e(a) = 0$. ■

Proof of Theorem 2.3. $(e, e) = \frac{1}{|G|} \sum |e(a)|^2 = 1$; for $e \neq e'$, $e\overline{e'}$ is a nontrivial character, so $(e, e') = 0$ by Lemma 2.4. ■

Hence $|\hat{G}| \leq |G|$.

7.2.4 Characters as a total family

Theorem 2.5

The characters of a finite abelian group G form a basis of the vector space V of functions on G; in particular $|\hat{G}| = |G|$.

Lemma 2.6

A commuting family $\{T_1, \ldots, T_k\}$ of unitary transformations of a finite-dimensional inner product space V ($T_i T_j = T_j T_i$) is simultaneously diagonalizable: V has a basis of common eigenvectors.

Proof. Induction on k; $k = 1$ is the spectral theorem for unitary operators. Write $V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_s}$ (eigenspaces of T_k). Each T_j, $j < k$, preserves V_{λ_i}: $T_k T_j v = T_j T_k v = \lambda_i T_j v$. By induction each V_{λ_i} has a basis of common eigenvectors of the restrictions; take the union. ■

Proof of Theorem 2.5. Let $(T_a f)(x) = f(ax)$: commuting (G abelian) and unitary ($x \mapsto ax$ permutes G). Lemma 2.6 gives a basis of common eigenvectors v, $T_a v = \lambda_a v$. Then $v(1) \neq 0$, else $v(a) = (T_a v)(1) = \lambda_a v(1) = 0$ for all a. So $w = v/v(1)$ satisfies $w(x) = \lambda_x$ ($v(x) = (T_x v)(1)$), $w(ab) = \lambda_{ab} = \lambda_a \lambda_b$ (from $T_{ab} = T_a T_b$), and $w \neq 0$ (T_x invertible); by Lemma 2.2 w is a character. So characters span V, and with Theorem 2.3 form a basis. ■

7.2.5 Fourier inversion and Plancherel formula

For f on G and $e \in \hat{G}$: the **Fourier coefficient** $\hat{f}(e) = (f, e) = \frac{1}{|G|}\sum_a f(a)\overline{e(a)}$ and the **Fourier series** $f \sim \sum_{e\in\hat{G}} \hat{f}(e)e$.

Theorem 2.7

The characters of a finite abelian group G form an orthonormal basis of V with $(f, g) = \frac{1}{|G|}\sum_a f(a)\overline{g(a)}$. For every f on G, $f = \sum_{e\in\hat{G}} \hat{f}(e)e$ (inversion) and $\|f\|^2 = \sum_{e\in\hat{G}} |\hat{f}(e)|^2$ (Plancherel).

Proof. Theorems 2.3 and 2.5, then expansion in an orthonormal basis and Pythagoras. ∎

For $G = \mathbb{Z}(N)$ this is Theorem 1.2 (up to normalization).

7.3 Selected exercises

Motivation. Book exercises linking the finite theory to the circle, and finite Fourier series as convolution with a delta kernel.

Exercise 7.2 (Decay of discrete Fourier coefficients)

Let f be a C^1 function on the circle and, for $N \geq 1$, $a_N(n) = \frac{1}{N}\sum_{k=1}^{N} f(e^{2\pi ik/N})e^{-2\pi ikn/N}$. Prove that there is c, independent of N and n, with $|a_N(n)| \leq c/|n|$ whenever $0 < |n| \leq N/2$.

Solution. Let $g(x) = f(e^{2\pi ix})$ (1-periodic, C^1) and $M = \sup|g'|$.

Step 1. For integer ℓ, reindexing $k \mapsto k - \ell$ (N-periodic summand),

$$\frac{1}{N}\sum_{k=1}^{N} g\Big(\frac{k+\ell}{N}\Big)e^{-2\pi ikn/N} = e^{2\pi i\ell n/N}a_N(n).$$

Subtracting from the definition,

$$a_N(n)(1 - e^{2\pi i\ell n/N}) = \frac{1}{N}\sum_{k=1}^{N}\Big[g\Big(\frac{k}{N}\Big) - g\Big(\frac{k+\ell}{N}\Big)\Big]e^{-2\pi ikn/N}.$$

Step 2. By the mean value theorem the right side is at most $M\ell/N$.

Step 3. Let $\ell = \lfloor N/(2|n|)\rfloor \geq 1$. Then $\ell|n|/N \leq \frac{1}{2}$, and $\ell|n|/N \geq \max(|n|/N, \frac{1}{2} - |n|/N) \geq \frac{1}{4}$. So $s = \ell n/N$ has $\frac{1}{4} \leq |s| \leq \frac{1}{2}$ and

$$|1 - e^{2\pi is}| = 2|\sin(\pi s)| \geq 2\sin(\pi/4) = \sqrt{2}.$$

Step 4. So $|a_N(n)| \leq \frac{M\ell}{\sqrt{2}N} \leq \frac{M}{2\sqrt{2}\,|n|}$: take $c = M/(2\sqrt{2})$. ∎

Exercise 7.8 (Sampling a trigonometric polynomial)

Let $P(x) = \sum_{n=1}^{N} a_n e^{2\pi inx}$.
(a) Show that $\int_0^1 |P(x)|^2 dx = \frac{1}{N}\sum_{j=1}^{N} |P(j/N)|^2$.

(b) Prove that $P(x) = \sum_{j=1}^{N} P(j/N)K(x - j/N)$, where

$$K(x) = \frac{1}{N}\sum_{n=1}^{N} e^{2\pi i n x} = \frac{e^{2\pi i x}}{N}\,\frac{1 - e^{2\pi i N x}}{1 - e^{2\pi i x}} \qquad (x \notin \mathbb{Z}),$$

and that $K(0) = 1$, $K(j/N) = 0$ for $j \not\equiv 0 \pmod N$.

Solution. Let $\zeta = e^{2\pi i/N}$, so $P(j/N) = \sum_n a_n \zeta^{nj}$.

Step 1. For $n, m \in \{1, \dots, N\}$, $\frac{1}{N}\sum_{j=1}^{N} \zeta^{(n-m)j} = \mathbb{1}\{n = m\}$ by Lemma 1.1 ($j = N$ may replace $j = 0$, and $n - m \in (-N, N)$ is $\equiv 0$ only if $n = m$).

Step 2. So

$$\frac{1}{N}\sum_{j=1}^{N} P(j/N)\zeta^{-mj} = \sum_{n=1}^{N} a_n \frac{1}{N}\sum_{j=1}^{N} \zeta^{(n-m)j} = a_m, \qquad 1 \le m \le N.$$

(a) $\int_0^1 e^{2\pi i(n-m)x}dx = \mathbb{1}\{n = m\}$ gives $\int_0^1 |P|^2 = \sum_n |a_n|^2$, and by Step 1

$$\frac{1}{N}\sum_{j=1}^{N} |P(j/N)|^2 = \sum_{n,m} a_n \bar{a}_m \frac{1}{N}\sum_{j=1}^{N} \zeta^{(n-m)j} = \sum_n |a_n|^2.$$

(Parseval on the circle and, by Step 2, on $\mathbb{Z}(N)$, Theorem 1.2.)

(b) Insert Step 2 into P:

$$P(x) = \sum_{n=1}^{N} \Big(\frac{1}{N}\sum_{j=1}^{N} P(j/N)e^{-2\pi i n j/N}\Big)e^{2\pi i n x} = \sum_{j=1}^{N} P(j/N)\,\frac{1}{N}\sum_{n=1}^{N} e^{2\pi i n(x - j/N)},$$

i.e. $\sum_j P(j/N)K(x - j/N)$. For $x \notin \mathbb{Z}$, $q = e^{2\pi i x} \neq 1$ and $\sum_{n=1}^{N} q^n = q\frac{1-q^N}{1-q}$. $K(0) = 1$, and for $j \not\equiv 0$, $q = \zeta^j \neq 1$, $q^N = 1$, so $K(j/N) = \frac{q}{N}\frac{1-q^N}{1-q} = 0$. So P is determined by its N samples. ∎

Exercise 7.13 (Convolution and the delta kernel on G)

Let G be a finite abelian group with unit 1_G and $(f * g)(a) = \frac{1}{|G|}\sum_{b\in G} f(b)g(ab^{-1})$.

(a) Show that $\widehat{f * g}(e) = \hat{f}(e)\hat{g}(e)$ for all $e \in \hat{G}$.

(b) Show that $\sum_{e\in\hat{G}} e(c) = 0$ whenever $c \neq 1_G$.

(c) Let $D(c) = \sum_{e\in\hat{G}} e(c)$. Show that the Fourier series $Sf = \sum_e \hat{f}(e)e$ equals $f * D$, and deduce $Sf = f$.

Solution.

(a) With $\overline{e(bc)} = \overline{e(b)}\,\overline{e(c)}$ and $c = ab^{-1}$ running over G,

$$\widehat{f * g}(e) = \frac{1}{|G|^2}\sum_{b\in G} f(b)\sum_{a\in G} g(ab^{-1})\overline{e(a)} = \frac{1}{|G|^2}\sum_{b\in G} f(b)\sum_{c\in G} g(c)\overline{e(bc)}$$
$$= \Big(\frac{1}{|G|}\sum_b f(b)\overline{e(b)}\Big)\Big(\frac{1}{|G|}\sum_c g(c)\overline{e(c)}\Big) = \hat{f}(e)\hat{g}(e).$$

(b)

Step 1. Some $e_0 \in \hat{G}$ has $e_0(c) \neq 1$: otherwise all characters, hence all functions (Theorem 2.5), agree at c and 1_G, false for the indicator of $\{1_G\}$.

Step 2. $e \mapsto e_0 e$ is a bijection of $\hat{G}$ (Lemma 2.1), so $\sum_e e(c) = e_0(c)\sum_e e(c)$, forcing $\sum_e e(c) = 0$.

(c)

Step 1. By (b) and $|\hat{G}| = |G|$ (Theorem 2.5), $D(1_G) = |G|$, $D(c) = 0$ for $c \neq 1_G$.

Step 2. With $\overline{e(b)} = e(b^{-1})$,

$$Sf(a) = \sum_{e \in \hat{G}} \frac{1}{|G|} \sum_{b \in G} f(b)\overline{e(b)}e(a) = \frac{1}{|G|} \sum_{b \in G} f(b) \sum_{e \in \hat{G}} e(ab^{-1}) = (f * D)(a).$$

Step 3. By Step 1 only $b = a$ contributes: $(f * D)(a) = f(a)$. So $Sf = f$, a second proof of inversion (Theorem 2.7). ■